

## Thermal Field Products

Ping Zhou Consulting · European Distribution

**Note:**

Values are based on manufacturer specifications. Custom dimensions and tolerances on request. Contact Ping Zhou Consulting for quotations and delivery within Europe.

Hinweis: Werte basieren auf Herstellerangaben. Sondermaße und Toleranzen auf Anfrage. Für Angebote und Lieferungen in Europa wenden Sie sich an Ping Zhou Consulting

## Graphite Soft Felt (Soft felt)

Graphit-Weichfilz



### Product Overview / Produktübersicht

|                                      |                                                                    |
|--------------------------------------|--------------------------------------------------------------------|
| <b>Product / Produkt</b>             | Graphite Soft Felt (Soft felt)                                     |
| <b>Specification / Spezifikation</b> | Thickness 5 / 10 mm · Width 1.3–1.6 m (custom on request)          |
| <b>Manufacturer / Hersteller</b>     | Made in China                                                      |
| <b>EU Distributor / EU-Vertrieb</b>  | Ping Zhou Consulting, Germany                                      |
| <b>Application / Anwendung</b>       | Flexible high-temperature insulation for thermal field assemblies. |
| <b>Anwendung (DE)</b>                | Flexible Hochtemperatur-Isolation für Thermofeld-Baugruppen.       |

### Technical Parameters / Technische Parameter

| Parameter / Parameter                                  | Unit / Einheit    | VLG-5 / VLG-10 | VLGY-5 / VLGY-10 |
|--------------------------------------------------------|-------------------|----------------|------------------|
| Thickness                                              | mm                | 5 / 10         | 5 / 10           |
| Density                                                | g/cm <sup>3</sup> | 0.08–0.11      | 0.07–0.09        |
| Width                                                  | m                 | 1.3–1.6        | 1.3–1.6          |
| Carbon content                                         | wt. %             | ≥ 99.80        | ≥ 99.80          |
| Ash content                                            | ppm               | ≤ 300          | ≤ 300            |
| Breaking strength                                      | MPa               | ≥ 0.3          | ≥ 0.2            |
| Thermal conductivity @ 1000 °C                         | W/m·K             | ≤ 0.17         | ≤ 0.23           |
| Thermal conductivity @ 1500 °C                         | W/m·K             | ≤ 0.35         | ≤ 0.38           |
| Processing temperature                                 | °C                | ≥ 1800         | ≥ 1800           |
| Fiber shedding rate                                    | %                 | ≤ 0.3          | ≤ 0.3            |
| Specific surface area                                  | m <sup>2</sup> /g | 3–7            | 3–7              |
| Maximum service temperature (non-oxidizing atmosphere) | °C                | ≤ 3200         | ≤ 3200           |

\*Processing temperature: Graphitization temperature >1800°C (in inert atmosphere/vacuum during manufacturing)

## Integral Needle-Punched Solidified Felt (Graphit Felt)

Integrale nadelfixierte Festkohlefaser-Matte



### Product Overview / Produktübersicht

|                                      |                                                                    |
|--------------------------------------|--------------------------------------------------------------------|
| <b>Product / Produkt</b>             | Integral Needle-Punched Solidified Felt                            |
| <b>Specification / Spezifikation</b> | 1500 mm × 1000 mm × 40 mm                                          |
| <b>Manufacturer / Hersteller</b>     | Made in China                                                      |
| <b>EU Distributor / EU-Vertrieb</b>  | Ping Zhou Consulting, Germany                                      |
| <b>Application / Anwendung</b>       | Thermal field insulation for high-temperature industrial furnaces. |
| <b>Anwendung (DE)</b>                | Wärmedämmung für Thermofelder in Hochtemperatur-Industrieöfen.     |

### Technical Parameters / Technische Parameter

| Parameter / Parameter                                  | Unit / Einheit       | Value / Wert                           |
|--------------------------------------------------------|----------------------|----------------------------------------|
| Density                                                | g/cm <sup>3</sup>    | 0.17-0.20                              |
| Carbon content                                         | wt. %                | > 99%                                  |
| Ash content                                            | ppm                  | ≤300                                   |
| Compressive strength                                   | Mpa                  | ≥2                                     |
| Flexural strength                                      | Mpa                  | ≥1                                     |
| Thermal conductivity (25 °C)                           | W/m·k                | ≤0.16                                  |
| Processing temperature                                 | °C                   | ≥1800                                  |
| Moisture absorption                                    | %                    | < 1                                    |
| Thickness                                              | mm                   | 5-200                                  |
| Maximum service temperature (non-oxidizing atmosphere) | °C                   | ≤3200                                  |
| Thermal expansion coefficient                          | ×10 <sup>-6</sup> /K | ≤0.25 (0-1000°C)<br>≤0.4 (1000-2000°C) |
| Machining accuracy                                     | mm                   | ±2                                     |

\*Processing temperature: Graphitization temperature >1800°C (in inert atmosphere/vacuum during manufacturing)

## Carbon Cloth Laminated Carbon Plate (CFC Plate)

Kohlefaserstoff-laminierte Kohleplatte



### Product Overview / Produktübersicht

|                                      |                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------|
| <b>Product / Produkt</b>             | Carbon Cloth Laminated Carbon Plate                                         |
| <b>Specification / Spezifikation</b> | 1000 mm × 1000 mm × 2 mm                                                    |
| <b>Manufacturer / Hersteller</b>     | Made in China                                                               |
| <b>EU Distributor / EU-Vertrieb</b>  | Ping Zhou Consulting, Germany                                               |
| <b>Application / Anwendung</b>       | Rigid carbon composite plate for thermal field and structural applications. |
| <b>Anwendung (DE)</b>                | Starre Kohlenstoffverbundplatte für Thermofeld- und Strukturanwendungen.    |

### Technical Parameters / Technische Parameter

| Parameter / Parameter                                  | Unit / Einheit       | Value / Wert                          |
|--------------------------------------------------------|----------------------|---------------------------------------|
| Process                                                | -                    | carbon cloth lamination               |
| Density                                                | g/cm <sup>3</sup>    | 1.20-1.50                             |
| Carbon content                                         | wt. %                | ≥99.5                                 |
| Ash content                                            | ppm                  | ≤300                                  |
| Compressive strength                                   | Mpa                  | ≥150                                  |
| Flexural strength                                      | Mpa                  | ≥110                                  |
| Thermal conductivity (25°C)                            | W/m·k                | 26 ± 3                                |
| Processing temperature*                                | °C                   | ≥1800                                 |
| Moisture absorption                                    | %                    | < 1                                   |
| Thickness                                              | mm                   | 1.0-50                                |
| Maximum service temperature (non-oxidizing atmosphere) | °C                   | ≤3200                                 |
| Thermal expansion coefficient                          | ×10 <sup>-6</sup> /K | ≤0.2 (0-1000°C)<br>≤0.3 (1000-2000°C) |
| Machining accuracy                                     | mm                   | ±1                                    |

\*Processing temperature: Graphitization temperature >1800°C (in inert atmosphere/vacuum during manufacturing)

## U-Shaped Carbon-Carbon Edge (CFC U-Profile)

U-förmige Kohlenstoff-Kohlenstoff-Kante



### Product Overview / Produktübersicht

|                                      |                                                                          |
|--------------------------------------|--------------------------------------------------------------------------|
| <b>Product / Produkt</b>             | U-Shaped Carbon-Carbon Edge                                              |
| <b>Specification / Spezifikation</b> | 1000 mm × 1000 mm × 2 mm                                                 |
| <b>Manufacturer / Hersteller</b>     | Made in China                                                            |
| <b>EU Distributor / EU-Vertrieb</b>  | Ping Zhou Consulting, Germany                                            |
| <b>Application / Anwendung</b>       | Edge protection and thermal insulation component for furnace assemblies. |
| <b>Anwendung (DE)</b>                | Kantenschutz- und Wärmedämmkomponente für Ofenbaugruppen.                |

### Technical Parameters / Technische Parameter

| Parameter / Parameter                                  | Unit / Einheit       | Value / Wert                          |
|--------------------------------------------------------|----------------------|---------------------------------------|
| Process                                                | -                    | Carbon cloth lamination               |
| Density                                                | g/cm <sup>3</sup>    | 1.20-1.40                             |
| Carbon content                                         | wt. %                | ≥99.5                                 |
| Ash content                                            | ppm                  | ≤300                                  |
| Compressive strength                                   | Mpa                  | ≥120                                  |
| Flexural strength                                      | Mpa                  | ≥100                                  |
| Thermal conductivity (25°C)                            | W/m·K                | ≤30 ⊥ ≤5                              |
| Processing temperature*                                | °C                   | ≥1800                                 |
| Moisture absorption                                    | %                    | < 1                                   |
| Thickness                                              | mm                   | 1.0-2.0                               |
| Maximum service temperature (non-oxidizing atmosphere) | °C                   | ≤3200                                 |
| Thermal expansion coefficient                          | ×10 <sup>-6</sup> /K | ≤0.2 (0-1000°C)<br>≤0.3 (1000-2000°C) |
| Machining accuracy                                     | mm                   | ±1                                    |

\*Processing temperature: Graphitization temperature >1800°C (in inert atmosphere/vacuum during manufacturing)